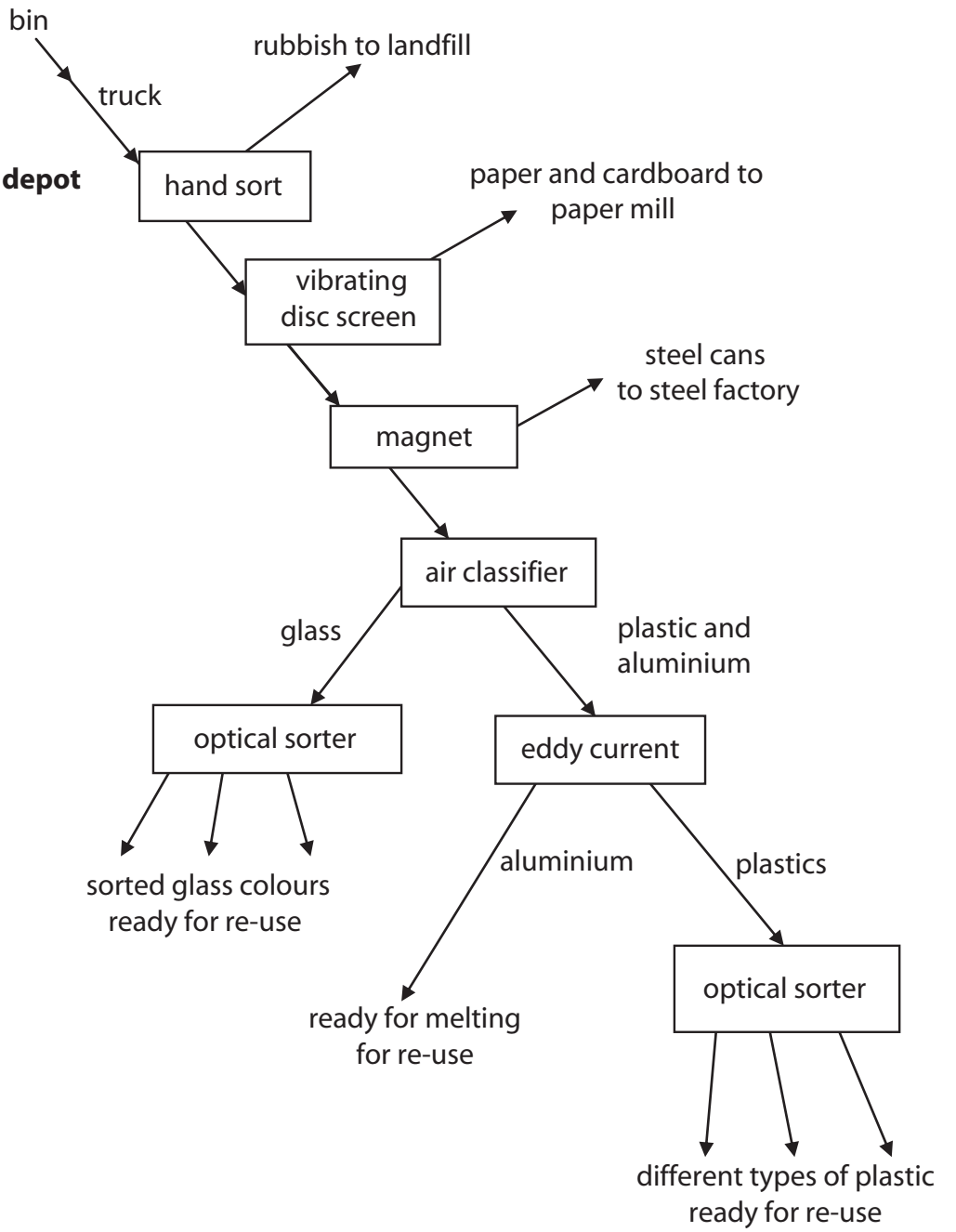
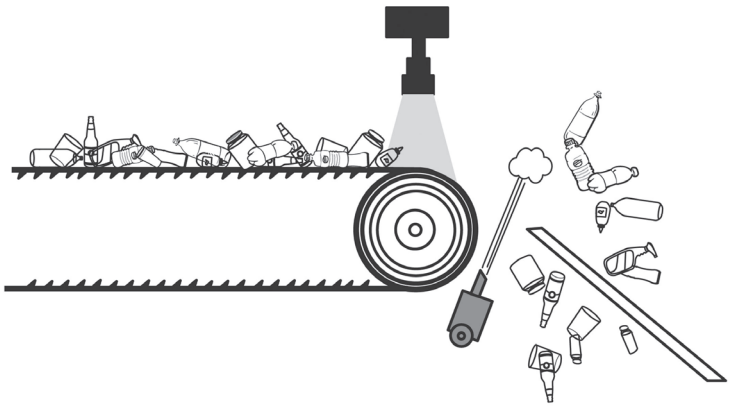
Recycling separators

After your recycling bin is collected the rubbish needs to be sorted before it can be **recycled**. Municipal councils use systems like this for sorting recycled material.



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Optical sorting



After the air classifier has separated them, glass and plastic containers each go to their own **optical sorter**. The optical sorter shines a strong light on each container. A detector then decides what kind of plastic or glass it is, and as it falls, a blast of air directs it into the correct storage bin, ready to be recycled.

☞ **1** Use the word bank to complete the passage below.

Optical _____ machines use light to sort _____ and _____ into their different types and _____. Until they are sorted, they cannot be _____ into new containers. A beam of _____ shines on each piece and _____ back. Each _____ type of material reflects different colours. A _____ detector recognises each type and fires a blast of air at it to push it into the _____ storage bin.

After passing through the optical sorter, the different _____ of plastic or glass will be in _____ large bins, ready to go to the plant to be made into _____ again.

Word bank

computerised
reflects
plastics
recycled
separate
correct
light
sorting
colours
different
types
containers
glass

2 Plastic has two major advantages over glass. What are they?

_____

_____

3 Plastics that are not recyclable, such as polystyrene, have no recycling stamp on them. Have a look at a plastic bottle to see if it can be recycled. Draw the stamp that tells you that it can be.

4 What does the number in the stamp tell you? _____

_____

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